

1	Course Name:
	Algorithm Complexity Analysis
2	Course Code:
	IDB40403
3	Year and Semester Offered:
	Year 2 Semester 2
4	Credit Hours Value:
	3 Credit Hours [3 SLT Credit]
5	Prerequisite(s), if applicable:
	NIL
6	Course Synopsis:
	This course is related to the study of computational complexity of algorithms and data structures in computer science. Topics covered include the roles of algorithms in computing, functions' growth, divide-and-conquer, probabilistic analysis and randomized algorithms, sorting algorithms, tree data structures, dynamic programming, greedy algorithms, amortized analysis and graph algorithms.
7	Course Learning Outcome:
	Upon completion of the course, students should be able to: 1. Analyse asymptotic performance for some basic algorithmic examples. 2. Propose optimal solutions for many of the representative problems in data structures and analysis of algorithms. 3. Assess the efficiency of the algorithm in terms of time, and resource such as memory.
8	Content Outline:
	Topic
	1. Algorithms in Computing
	2. Sorting Algorithms
	3. Advanced Data Structures
	4. Dynamic Programming and Greedy Algorithm
	5. Graph algorithms
	6. Optimization algorithms
9	References:
	Christos H. Papadimitriou, Combinatorial Optimization: Algorithms and Complexity : Dover Publications, 2013 Jeff Erickson, Algorithms: Independently Published(2019)